

OAI 5G RF Simulator — Quick Daily Workflow

This is the minimal set of commands you need every time you sit down to work.

1. Go to the OAI Deployment Directory

```
cd ~/OAI_Beginners_Experiment/openairinterface5g/ci-scripts/yaml_files/5g_rfsimulator
```

Confirm the files exist:

```
ls
```

You should see:

```
docker-compose.yaml  
oai_db.sql  
mini_nonrf_config.yaml
```

2. Check Docker Service

Verify Docker is running:

```
systemctl status docker
```

If not running:

```
sudo systemctl start docker
```

3. Check Existing Containers

See all OAI containers:

```
docker compose ps -a
```

or

```
docker ps -a
```

Look for:

```
rfsim5g-mysql  
rfsim5g-oai-amf
```

```
rfsim5g-oai-smf  
rfsim5g-oai-upf  
rfsim5g-oai-ext-dn  
rfsim5g-oai-gnb  
rfsim5g-oai-nr-ue
```

4. Start Everything (If Not Running)

If containers are stopped:

```
docker compose up -d
```

Wait about 30-60 seconds. Check status:

```
docker compose ps -a
```

Expected:

```
healthy  
healthy  
healthy
```

for MySQL, AMF, SMF, and UPF.

5. Check gNB Connection to AMF

```
docker logs rfsim5g-oai-amf 2>&1 | grep -A 6 "gNBs"
```

Expected:

```
Connected  
gnb-rfsim  
208,99
```

6. Check UE Registration

```
docker logs rfsim5g-oai-amf 2>&1 | grep "5GMM"
```

or

```
docker logs rfsim5g-oai-amf 2>&1 | grep "REGISTERED"
```

Success indicator:

```
5GMM-REGISTERED
```

7. Enter the UE Container

```
docker exec -it rfsim5g-oai-nr-ue bash
```

Inside the container:

```
ip addr
```

Look for oaitun_ue1 with an IP like 12.1.1.x.

Exit:

```
exit
```

8. Test UE Connectivity

```
docker exec -it rfsim5g-oai-nr-ue ping -I oaitun_ue1 -c 5 192.168.72.135
```

Expected:

```
0% packet loss
```

9. View Important Logs

AMF Logs:

```
docker logs rfsim5g-oai-amf | tail -50
```

gNB Logs:

```
docker logs rfsim5g-oai-gnb | tail -50
```

UE Logs:

```
docker logs rfsim5g-oai-nr-ue | tail -50
```

10. Check Current UE Identity

```
cd ~/OAI_Begineers_Experiment/openairinterface5g
grep -E "imsi|key|opc" ci-scripts/conf_files/nrue.uicc.conf
```

This shows the IMSI, Key, and OPc used by the software UE.

11. Check Subscriber Exists in Database

```
grep -i "208990100001100" ci-scripts/yaml_files/5g_rfsimulator/oai_db.sql
```

This verifies the default subscriber exists.

12. Restart Only the UE

After editing `nrue.uicc.conf`:

```
cd ci-scripts/yaml_files/5g_rfsimulator
docker compose restart oai-nr-ue
```

Verify registration again:

```
docker logs rfsim5g-oai-amf 2>&1 | grep "REGISTERED"
```

13. Stop Everything

```
docker compose down
```

 *This keeps the database.*

14. Destroy Everything and Recreate from Scratch

```
docker compose down -v
```

 *This deletes the MySQL volume and reloads subscribers from `oai_db.sql`.*

Only use this when you intentionally want a fresh deployment.